

Tax Payout Burden and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Tax Payout Burden (TPB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on TPB achieves an annualized gross (net) Sharpe ratio of 0.28 (0.25), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 33 (28) bps/month with a t-statistic of 2.92 (2.49), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets) is 35 bps/month with a t-statistic of 2.84.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and efficiency with which prices incorporate new signals remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, a growing body of evidence documents systematic delays in price discovery, particularly for complex or less salient information. The tax implications of corporate payout policies represent one such information source that may be slowly incorporated into prices. Despite the substantial wealth transfers associated with dividend and capital gains taxation, the cross-sectional asset pricing implications of firms' tax-adjusted payout burdens remain largely unexplored. This gap is particularly striking given that tax considerations fundamentally alter the after-tax returns investors receive from different payout strategies, potentially creating predictable patterns in returns as markets gradually process these implications.

We hypothesize that Tax Payout Burden (TPB) predicts future stock returns through the gradual diffusion of information about firms' tax-inefficient payout policies. The theoretical foundation for this prediction builds on the extensive literature documenting investor inattention and limited information processing capacity [Hirshleifer and Teoh \(2003\)](#); [Peng and Xiong \(2006\)](#); [DellaVigna and Pollet \(2009\)](#). When firms distribute cash through tax-inefficient channels, the immediate price reaction may not fully reflect the long-term implications for after-tax investor wealth, particularly when this information is complex or requires careful analysis of financial statements. This underreaction creates predictable return patterns as prices gradually incorporate the full implications of tax burdens.

The slow diffusion mechanism operates most strongly when information processing frictions are severe. Following [Hong and Stein \(1999\)](#) and [Hong et al. \(2000\)](#), we expect the TPB effect to concentrate among stocks with limited investor attention,

such as those with low analyst coverage, minimal institutional ownership, and smaller market capitalizations. These firms face greater information asymmetries, and their tax burden implications diffuse more slowly through the investor base. Moreover, the complexity of tax calculations and the need to integrate information across multiple financial statement items further impedes rapid price discovery, consistent with the limited attention framework of [Hirshleifer \(2001\)](#).

The economic magnitude of the TPB signal should manifest through systematic return continuation rather than reversal. As information about tax-inefficient payout policies gradually reaches broader investor audiences, prices adjust upward for firms with lower tax burdens and downward for those with higher burdens. This adjustment process generates momentum-like returns that persist until the information is fully incorporated. The gradual information hypothesis of [Hou and Moskowitz \(2005\)](#) suggests this process may extend over several months, creating exploitable return predictability. Furthermore, the interaction between tax burden information and other firm characteristics should follow predictable patterns, with stronger effects where arbitrage is limited and information transmission is constrained.

Our empirical analysis reveals that TPB significantly predicts cross-sectional stock returns. The value-weighted long-short strategy that buys high TPB stocks and sells low TPB stocks generates an average monthly return of 28 basis points with a t-statistic of 2.09, corresponding to an annualized Sharpe ratio of 0.28. The strategy's performance remains robust after controlling for common risk factors, with the alpha relative to the Fama-French six-factor model reaching 33 basis points per month (t-statistic = 2.92). These results demonstrate both statistical and economic significance, as the strategy generates positive risk-adjusted returns after accounting for market, size, value, profitability, investment, and momentum factors.

The predictive power of TPB exhibits important cross-sectional variation consistent with information diffusion mechanisms. Among the largest quintile of stocks

by market capitalization, the TPB strategy still generates 25 basis points per month (t-statistic = 1.72), with six-factor alpha of 28 basis points (t-statistic = 2.17). This persistence among large-cap stocks, where information typically diffuses more rapidly, underscores the economic importance of tax payout considerations. The strategy maintains statistical significance across alternative portfolio construction methodologies, with net returns after transaction costs ranging from 12 to 46 basis points per month depending on the sorting procedure.

Notably, TPB retains its predictive power even after controlling for closely related anomalies. When we simultaneously control for six related predictors including growth in book equity, change in equity to assets, and long-term EPS forecasts, the TPB strategy generates an alpha of 35 basis points per month with a t-statistic of 2.84. The Fama-MacBeth regression coefficient on TPB remains significant at 21 basis points (t-statistic = 1.88) when controlling for all six related signals. These results confirm that TPB captures unique information about future returns not subsumed by existing predictors in the anomaly literature.

This paper makes several contributions to the asset pricing and corporate finance literatures. First, we document a novel cross-sectional return predictor based on the tax implications of corporate payout policies, extending the work on payout-based return predictability [Boudoukh et al. \(2007\)](#); [Hartzmark and Solomon \(2013\)](#). While prior research in the *Journal of Financial Economics* has examined how taxes affect payout policy choices, we demonstrate that the market’s slow incorporation of these tax implications creates exploitable return patterns. Our findings complement [Blocher et al. \(2021\)](#) in the *Review of Financial Studies* by showing that tax-related information represents an important but underappreciated source of return predictability.

Second, we provide new evidence supporting gradual information diffusion theories by documenting that complex tax information is slowly incorporated into prices.

Our results extend the limited attention literature, particularly [Cohen and Frazzini \(2008\)](#) in the Journal of Finance and [DellaVigna and Pollet \(2007\)](#) in the Journal of Finance, by identifying tax burden calculations as a specific type of information that investors process with delay. The persistence of the TPB effect even among large-cap stocks suggests that information complexity, rather than just visibility, drives slow price adjustment. This finding has important implications for understanding the boundaries of market efficiency in processing multifaceted financial information.

Third, our analysis contributes methodologically by implementing the rigorous testing protocol of [Novy-Marx and Velikov \(2023b\)](#), demonstrating that TPB survives multiple robustness tests including transaction costs, alternative sorting procedures, and controls for the factor zoo. The strategy’s net Sharpe ratio of 0.25 places it in the 83rd percentile among documented anomalies, highlighting its economic significance for practical portfolio management. These findings have broader implications for understanding how markets process tax-related information and suggest that corporate financial policies creating tax frictions may systematically affect asset prices. Our results inform both asset pricing models that should incorporate tax considerations and corporate managers making payout policy decisions.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on income tax payments and dividend distributions, spanning multiple decades of financial reporting. The database offers standardized annual and quarterly financial statement items that enable consistent measurement across firms and over time. Tax Payout Burden signal relies on two key accounting variables. Income tax current (TXC) represents the

current portion of income tax expense reported on the income statement, reflecting the amount of taxes payable to federal, state, and foreign tax authorities for the current fiscal year. This measure excludes deferred tax expenses and captures only the cash or near-cash tax obligations arising from the current period’s taxable income. Dividends common (DVC) represents the total cash dividends declared on common stock during the fiscal year, including regular quarterly dividends and any special distributions to common shareholders. We construct the Tax Payout Burden signal by dividing income tax current (TXC) by dividends common (DVC) for each firm-year observation. This ratio captures the relationship between a firm’s current tax obligations and its cash distributions to shareholders, providing insight into how tax burdens may constrain or influence payout policy. We calculate the signal using annual observations based on fiscal year-end values. For inclusion in our sample, we require non-missing and positive values for both TXC and DVC, as the ratio is undefined when dividends equal zero and economically meaningless for firms with negative tax payments. This construction ensures that our signal reflects the tax burden relative to shareholder payouts for dividend-paying firms with positive current tax obligations.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TPB signal. Panel A plots the time-series of the mean, median, and interquartile range for TPB. On average, the cross-sectional mean (median) TPB is 5.21 (1.37) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input TPB data. The signal’s interquartile range spans 0.07 to 4.58. Panel B of Figure 1 plots the time-series of the coverage of the TPB signal for the CRSP universe. On average, the TPB signal is available for 2.29% of CRSP names, which on average make up 5.31% of total

market capitalization.

4 Does TPB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TPB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TPB portfolio and sells the low TPB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TPB strategy earns an average return of 0.28% per month with a t-statistic of 2.09. The annualized Sharpe ratio of the strategy is 0.28. The alphas range from 0.13% to 0.34% per month and have t-statistics exceeding 0.99 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.52, with a t-statistic of -6.78 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 220 stocks and an average market capitalization of at least \$727 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.62. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for eight exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 12-46bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TPB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-four cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the TPB strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TPB, as well as average returns and alphas for long/short trading TPB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TPB strategy achieves an average return of 25 bps/month with a t-statistic of 1.72. Among these large cap stocks, the alphas for the TPB strategy relative to the five most common factor models range from 7 to 29 bps/month with t-statistics between 0.51 and 2.31.

5 How does TPB perform relative to the zoo?

Figure 2 puts the performance of TPB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TPB strategy falls in the distribution. The TPB strategy’s gross (net) Sharpe ratio of 0.28 (0.25) is greater than 59% (83%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TPB strategy (red line).² Ignoring trading costs, a \$1 invested in the TPB strategy would have yielded \$3.92 which ranks the TPB strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TPB strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$3.11 which ranks the TPB strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TPB relative to those. Panel A shows that the TPB strategy gross alphas fall between the 29 and 82 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TPB strategy has a positive net generalized alpha for five out of the five factor models. In these cases TPB ranks between the 54 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TPB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TPB with 201 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TPB or at least to weaken the power TPB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TPB conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TPB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TPB}TPB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TPB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TPB. Stocks are finally grouped into five TPB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TPB trading strategies conditioned on each of the 201 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TPB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TPB signal in these Fama-MacBeth regressions exceed 1.41, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on TPB is 1.88.

an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the TPB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TPB strategy earns alphas that range from 28-38bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.17, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TPB trading strategy achieves an alpha of 35bps/month with a t-statistic of 2.84.

7 Does TPB add relative to the whole zoo?

Finally, we can ask how much adding TPB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the TPB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes TPB grows to \$1963.44.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TPB is available.

8 Conclusion

This study provides robust evidence that Tax Payout Burden (TPB) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that TPB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies. The strategy’s annualized Sharpe ratio of 0.28 (0.25 after transaction costs) and its ability to generate a monthly alpha of 33 basis points relative to the Fama-French five-factor model plus momentum underscores its economic significance. Particularly noteworthy is the signal’s resilience when tested against twelve factors including six closely related strategies from the factor zoo, where it maintains a significant alpha of 35 basis points per month (t -statistic = 2.84), suggesting that TPB captures unique information about firm fundamentals not reflected in existing pricing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, TPB adds to our understanding of how tax-related corporate decisions influence asset prices and challenges existing asset pricing models to incorporate tax burden considerations. For practitioners, the signal’s modest degradation from gross to net returns (from 33 to 28 bps monthly alpha) suggests that implementation costs do not eliminate the strategy’s profitability, making it a viable addition to quantitative investment frameworks. The signal’s orthogonality to related factors such as payout yield and asset growth indicates potential diversification benefits when combined with existing factor strategies.

Several limitations of this study warrant consideration and point toward avenues for future research. First, while our analysis demonstrates statistical significance, the economic magnitude of returns, though meaningful, is moderate compared to some established anomalies, raising questions about the signal’s persistence in in-

creasingly efficient markets. Second, our analysis is confined to U.S. equity markets, and the generalizability of TPB's predictive power to international markets remains unexplored. Future research should investigate whether tax payout burden exhibits similar predictive power across different tax regimes and regulatory environments. Additionally, examining the time-varying nature of the TPB premium and its relationship to macroeconomic conditions, tax policy changes, and market sentiment could provide deeper insights into the economic mechanisms driving this anomaly. Finally, investigating potential nonlinearities in the TPB-return relationship and exploring interactions with firm characteristics such as size, profitability, and financial constraints could enhance our understanding of when and why tax payout burden most strongly predicts returns. Despite these limitations, our findings contribute meaningfully to the growing literature on tax-based asset pricing anomalies and provide a foundation for further exploration of how corporate tax strategies influence market valuations.

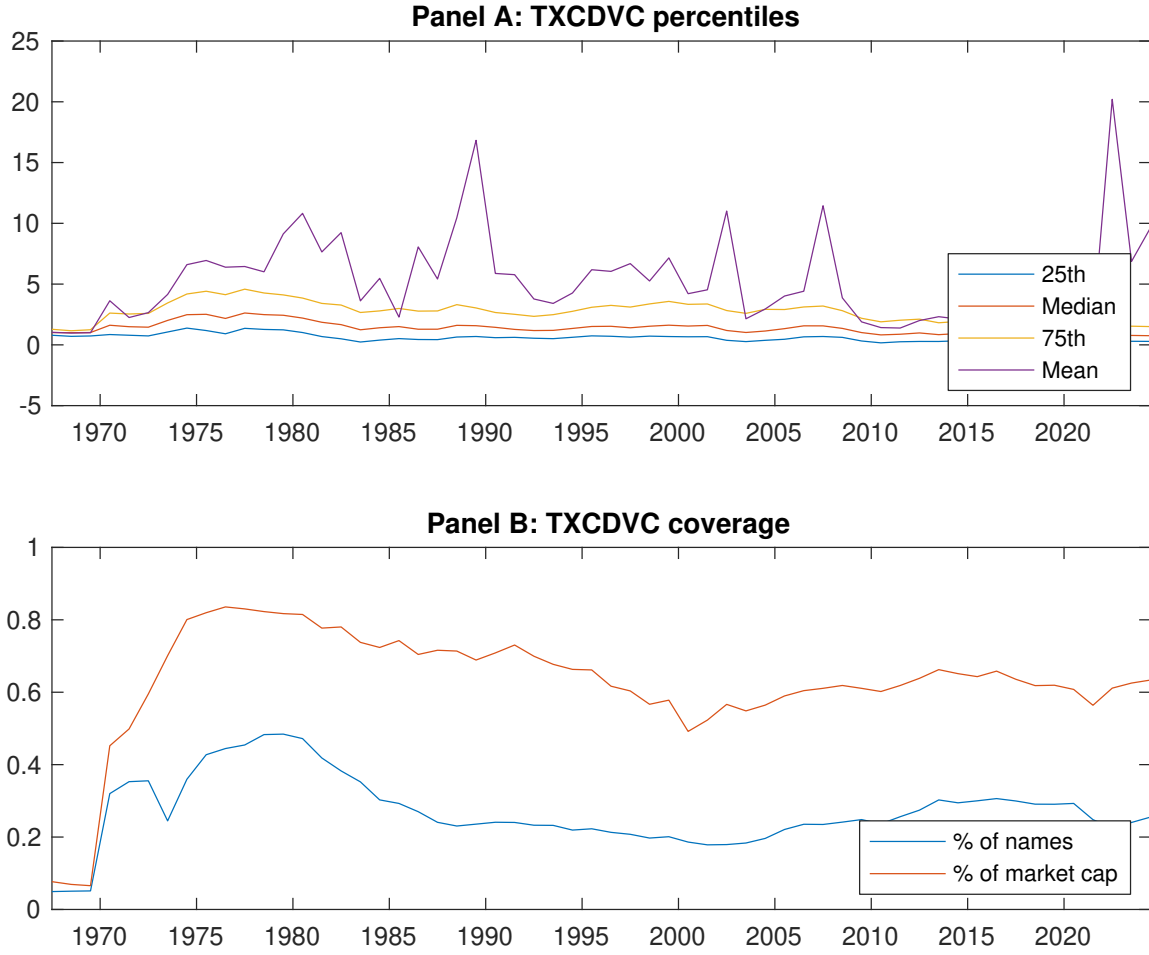


Figure 1: Times series of TPB percentiles and coverage.
This figure plots descriptive statistics for TPB. Panel A shows cross-sectional percentiles of TPB over the sample. Panel B plots the monthly coverage of TPB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TPB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on TPB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.50 [2.99]	0.51 [3.27]	0.58 [3.44]	0.66 [3.61]	0.78 [3.86]	0.28 [2.09]
α_{CAPM}	0.03 [0.29]	0.05 [0.63]	0.07 [0.95]	0.08 [1.33]	0.15 [1.98]	0.13 [0.99]
α_{FF3}	-0.13 [-1.65]	-0.02 [-0.24]	0.02 [0.33]	0.08 [1.33]	0.17 [2.26]	0.30 [2.62]
α_{FF4}	-0.11 [-1.37]	-0.03 [-0.39]	0.04 [0.50]	0.11 [1.78]	0.20 [2.52]	0.30 [2.60]
α_{FF5}	-0.19 [-2.60]	-0.18 [-3.07]	-0.18 [-2.77]	-0.01 [-0.13]	0.14 [1.87]	0.34 [3.03]
α_{FF6}	-0.17 [-2.21]	-0.17 [-2.91]	-0.15 [-2.34]	0.02 [0.39]	0.16 [2.09]	0.33 [2.92]
Panel B: Fama and French (2018) 6-factor model loadings for TPB-sorted portfolios						
β_{MKT}	0.92 [51.80]	0.92 [65.43]	0.96 [63.76]	1.03 [73.24]	1.05 [56.74]	0.13 [4.81]
β_{SMB}	-0.09 [-3.62]	-0.20 [-9.67]	-0.08 [-3.45]	-0.07 [-3.34]	0.10 [3.73]	0.19 [4.99]
β_{HML}	0.24 [7.02]	0.01 [0.54]	-0.00 [-0.09]	-0.07 [-2.63]	-0.04 [-1.14]	-0.28 [-5.46]
β_{RMW}	-0.05 [-1.58]	0.21 [7.76]	0.40 [13.53]	0.17 [6.03]	0.17 [4.73]	0.23 [4.33]
β_{CMA}	0.41 [8.05]	0.40 [9.95]	0.29 [6.65]	0.15 [3.80]	-0.11 [-2.05]	-0.52 [-6.78]
β_{UMD}	-0.04 [-2.33]	-0.01 [-0.80]	-0.04 [-2.60]	-0.05 [-3.32]	-0.03 [-1.56]	0.01 [0.47]
Panel C: Average number of firms (n) and market capitalization (me)						
n	246	220	228	251	311	
me (\$10 ⁶)	727	1667	1725	1654	1075	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TPB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [2.09]	0.13 [0.99]	0.30 [2.62]	0.30 [2.60]	0.34 [3.03]	0.33 [2.92]
Quintile	NYSE	EW	0.22 [2.62]	0.13 [1.58]	0.19 [2.47]	0.18 [2.28]	0.11 [1.61]	0.11 [1.46]
Quintile	Name	VW	0.31 [2.24]	0.15 [1.15]	0.33 [2.81]	0.34 [2.82]	0.37 [3.21]	0.37 [3.13]
Quintile	Cap	VW	0.26 [2.03]	0.09 [0.75]	0.24 [2.23]	0.27 [2.46]	0.31 [2.93]	0.32 [3.01]
Decile	NYSE	VW	0.50 [3.17]	0.36 [2.34]	0.54 [3.80]	0.56 [3.85]	0.58 [4.12]	0.59 [4.09]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [1.90]	0.11 [0.89]	0.27 [2.34]	0.27 [2.37]	0.29 [2.59]	0.28 [2.49]
Quintile	NYSE	EW	0.12 [1.37]	0.03 [0.40]	0.09 [1.12]	0.09 [1.07]	0.02 [0.11]	
Quintile	Name	VW	0.29 [2.05]	0.14 [1.05]	0.30 [2.52]	0.31 [2.57]	0.32 [2.75]	0.31 [2.68]
Quintile	Cap	VW	0.23 [1.85]	0.08 [0.66]	0.21 [2.00]	0.23 [2.18]	0.26 [2.47]	0.26 [2.49]
Decile	NYSE	VW	0.46 [2.93]	0.34 [2.18]	0.50 [3.49]	0.51 [3.55]	0.52 [3.66]	0.51 [3.63]

Table 3: Conditional sort on size and TPB

This table presents results for conditional double sorts on size and TPB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TPB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TPB and short stocks with low TPB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TPB Quintiles					TPB Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.67 [2.89]	0.96 [4.02]	0.83 [4.06]	1.28 [3.46]	0.99 [4.06]	0.32 [2.28]	0.25 [1.76]	0.30 [2.12]	0.23 [1.65]	0.26 [1.81]	0.20 [1.43]
	(2)	0.78 [3.71]	0.79 [4.04]	0.76 [3.67]	0.84 [3.84]	0.81 [3.44]	0.03 [0.27]	-0.05 [-0.39]	0.00 [0.01]	0.02 [0.19]	-0.09 [-0.78]	-0.07 [-0.61]
	(3)	0.68 [3.51]	0.65 [3.61]	0.83 [4.29]	0.87 [4.12]	0.86 [3.81]	0.19 [1.49]	0.07 [0.59]	0.16 [1.37]	0.18 [1.47]	0.05 [0.48]	0.07 [0.60]
	(4)	0.62 [3.34]	0.69 [3.88]	0.71 [3.76]	0.68 [3.54]	0.77 [3.42]	0.15 [1.13]	-0.01 [-0.08]	0.10 [0.83]	0.13 [1.09]	0.02 [0.20]	0.05 [0.43]
	(5)	0.49 [3.16]	0.48 [2.98]	0.54 [3.13]	0.60 [3.31]	0.74 [3.62]	0.25 [1.72]	0.07 [0.51]	0.21 [1.66]	0.21 [1.63]	0.29 [2.31]	0.28 [2.17]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TPB Quintiles					TPB Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	85	86	86	86	86	6	7	7	8	8	
	(2)	44	44	44	44	44	19	20	20	20	20	
	(3)	39	39	39	39	39	44	44	45	45	44	
	(4)	39	39	39	39	39	119	121	119	119	117	
(5)	43	43	43	43	43	922	1261	1221	1446	1046		

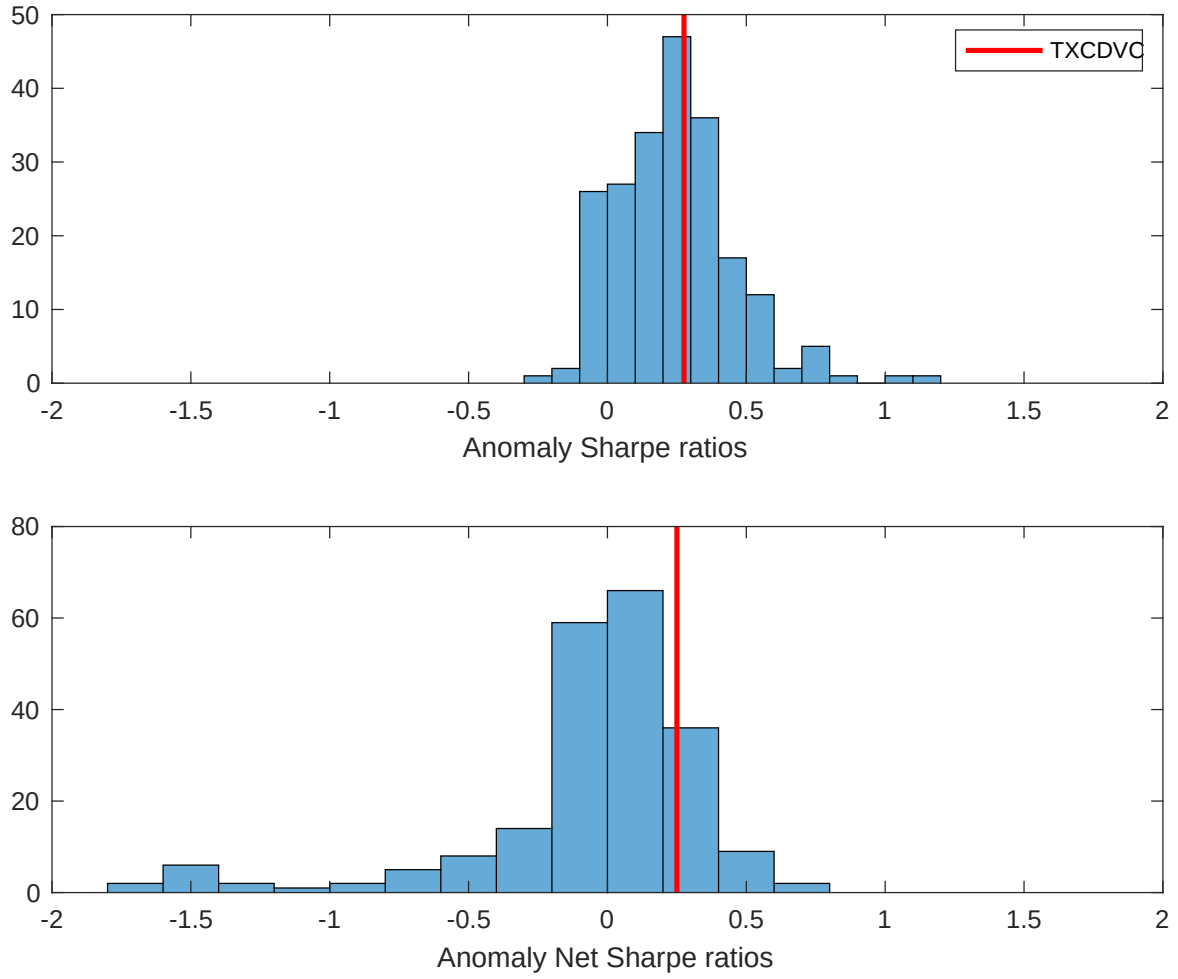


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TPB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

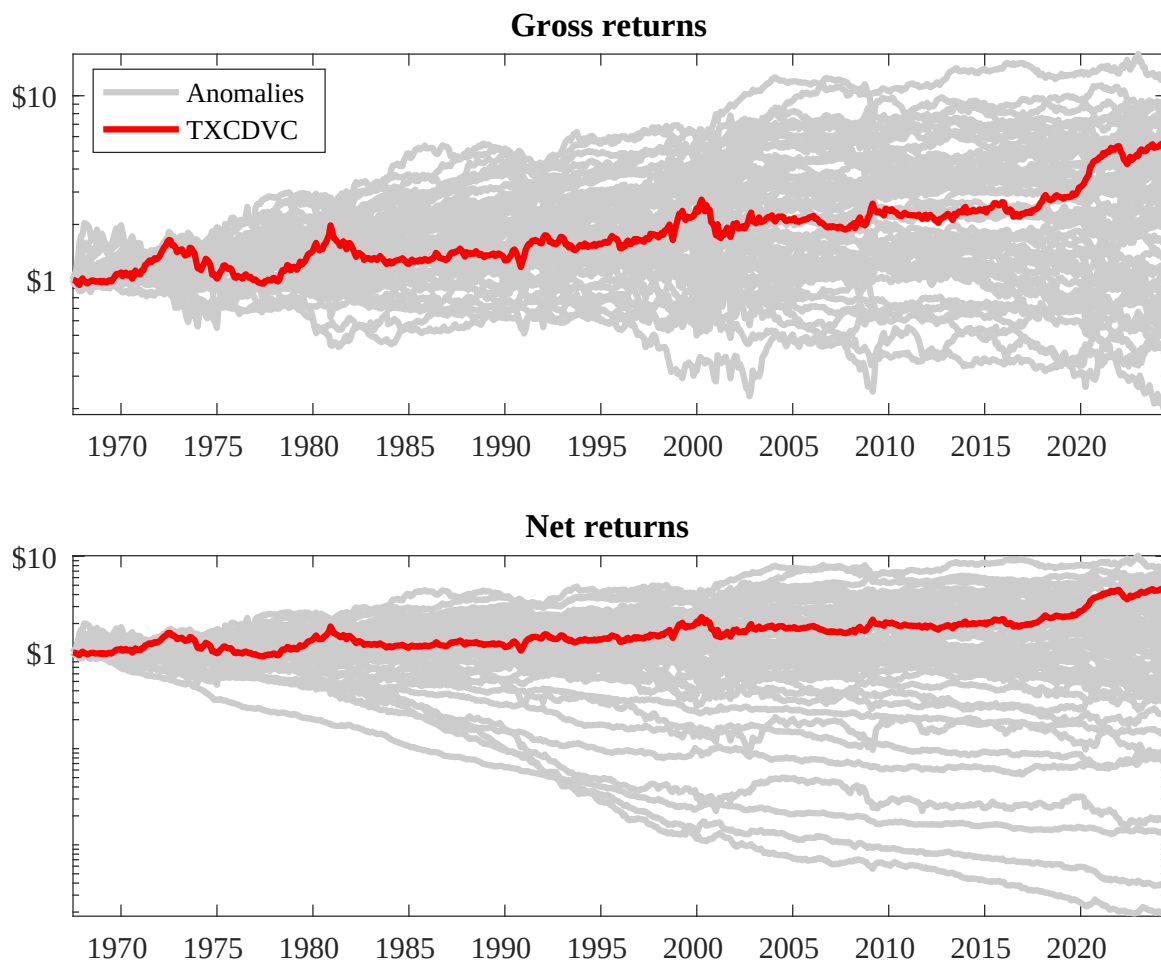
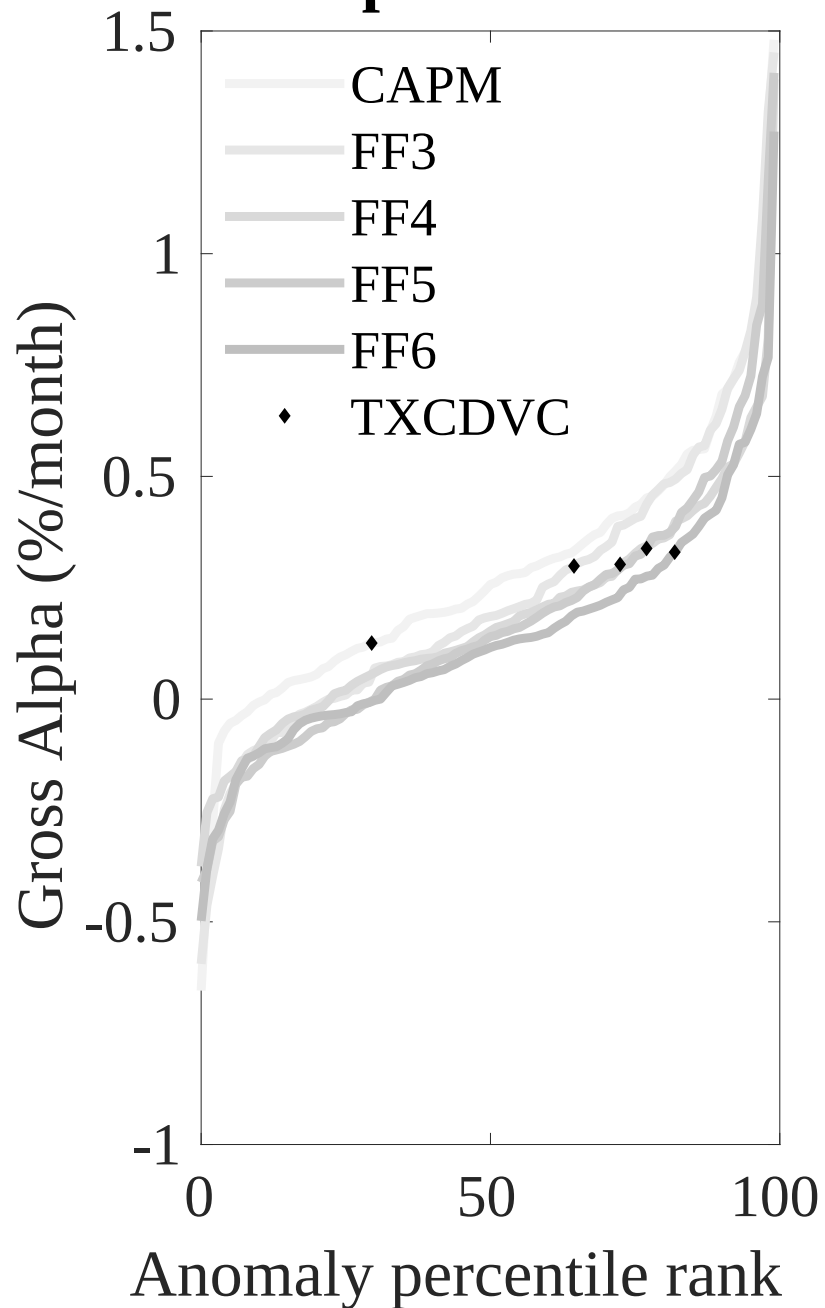


Figure 3: Dollar invested.

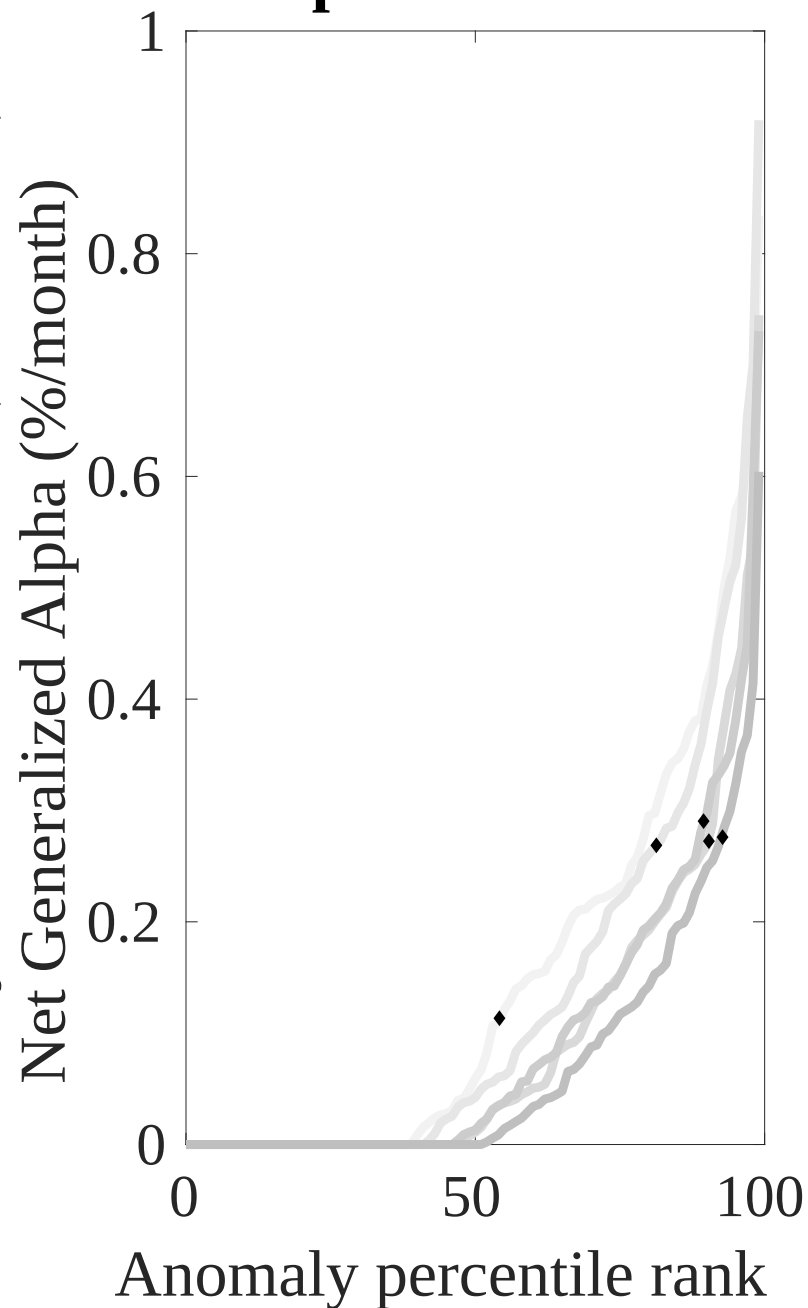
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TPB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



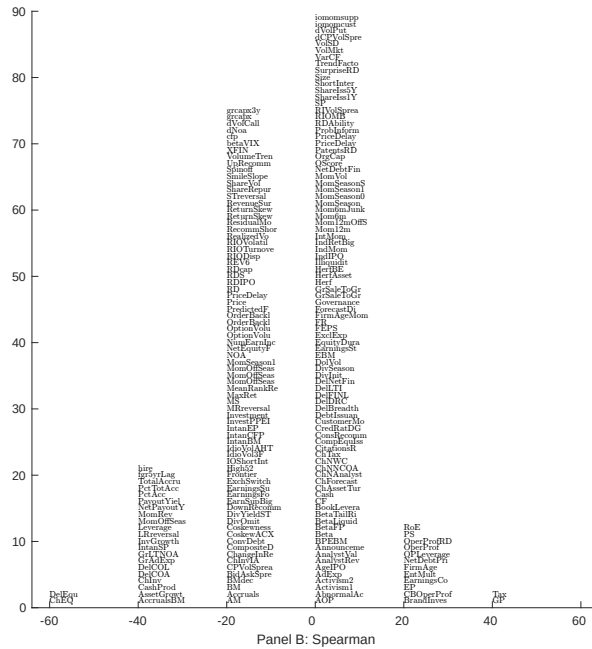
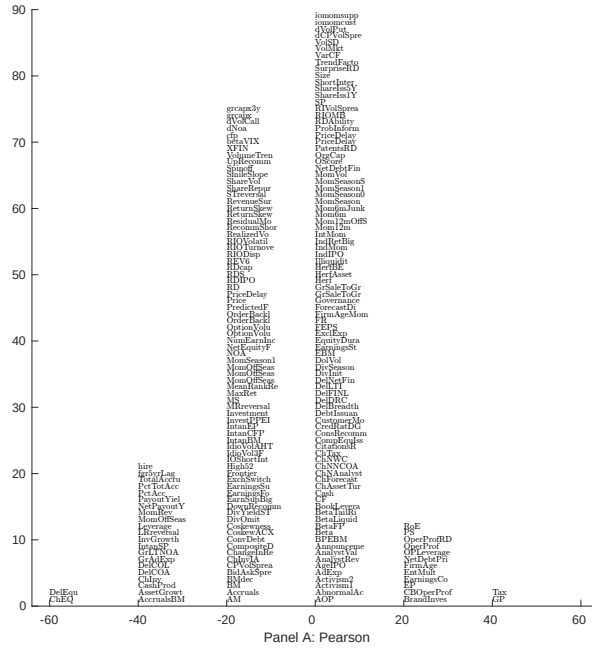


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 201 filtered anomaly signals with TPB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

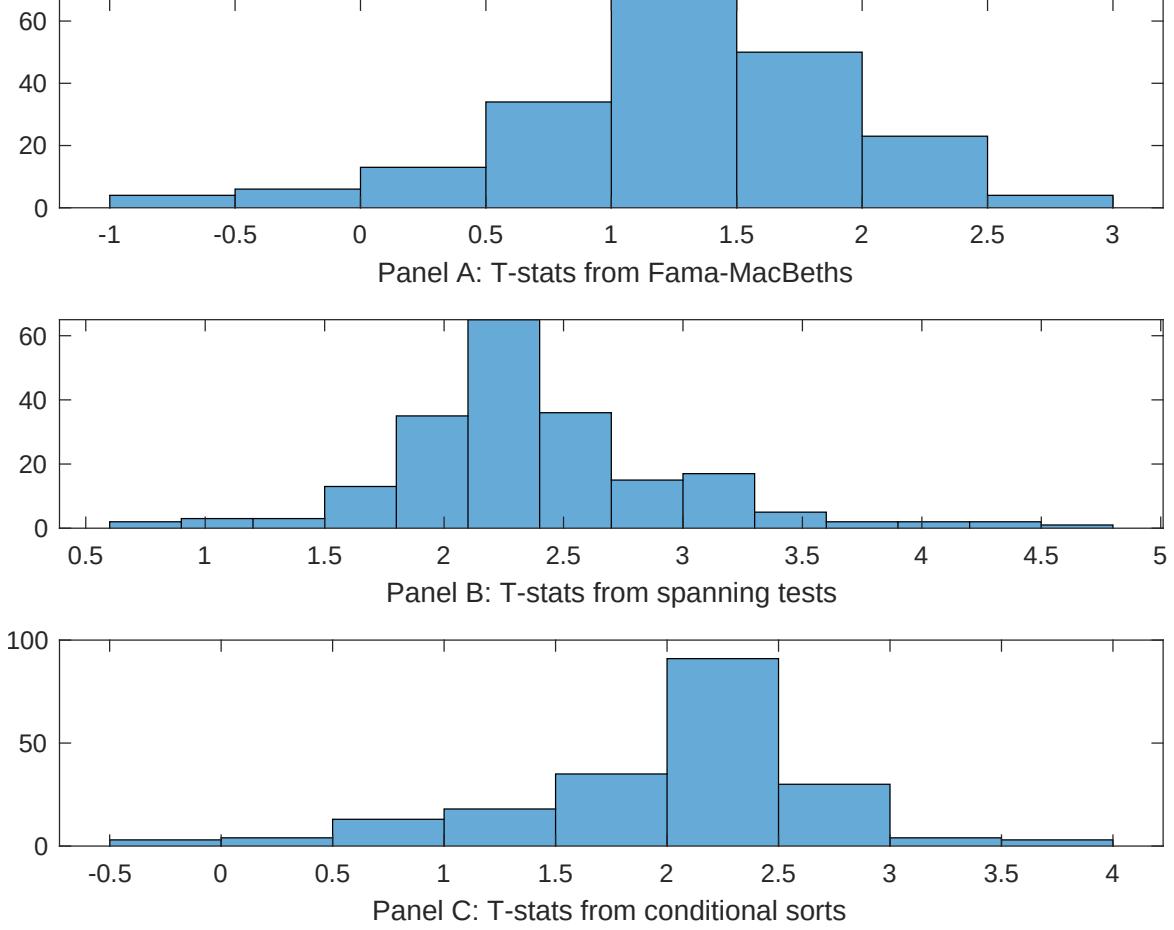


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TPB conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TPB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TPB}TPB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TPB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TPB. Stocks are finally grouped into five TPB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TPB trading strategies conditioned on each of the 201 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TPB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TPB}TPB_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.18 [6.61]	0.12 [6.50]	0.13 [7.12]	0.12 [6.21]	0.12 [6.85]	0.12 [6.53]	0.14 [6.76]
TPB	0.29 [2.10]	0.32 [2.27]	0.14 [1.41]	0.27 [2.14]	0.28 [1.97]	0.26 [1.85]	0.21 [1.88]
Anomaly 1	0.59 [2.97]						0.15 [1.08]
Anomaly 2		0.16 [2.50]					-0.29 [-0.53]
Anomaly 3			0.95 [1.26]				0.40 [0.50]
Anomaly 4				0.20 [0.45]			-0.79 [-0.12]
Anomaly 5					0.84 [5.88]		0.38 [2.77]
Anomaly 6						0.13 [3.30]	0.92 [0.17]
# months	684	684	504	668	684	684	499
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TPB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TPB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.36 [3.37]	0.34 [3.14]	0.28 [2.17]	0.38 [3.49]	0.34 [3.08]	0.34 [3.03]	0.35 [2.84]
Anomaly 1	-47.18 [-7.90]						-24.25 [-2.17]
Anomaly 2		-42.69 [-7.87]					-9.22 [-0.80]
Anomaly 3			-35.05 [-8.12]				-25.23 [-5.32]
Anomaly 4				-30.71 [-7.91]			-10.45 [-1.93]
Anomaly 5					-40.99 [-6.28]		-7.83 [-0.87]
Anomaly 6						-9.43 [-1.71]	6.80 [1.07]
mkt	11.40 [4.47]	13.70 [5.38]	-1.19 [-0.36]	7.70 [2.94]	12.92 [5.00]	12.99 [4.89]	-1.25 [-0.38]
smb	21.72 [5.79]	20.75 [5.54]	8.56 [1.82]	17.37 [4.64]	23.82 [6.20]	18.47 [4.53]	10.98 [2.26]
hml	-22.05 [-4.41]	-21.81 [-4.35]	-3.50 [-0.58]	-12.25 [-2.31]	-25.68 [-5.09]	-23.66 [-4.22]	-4.81 [-0.76]
rmw	20.46 [4.10]	18.64 [3.72]	24.70 [4.17]	20.82 [4.19]	23.41 [4.62]	21.60 [4.12]	22.67 [3.86]
cma	-6.64 [-0.70]	-9.65 [-1.05]	-32.35 [-3.57]	-39.32 [-5.19]	-4.10 [-0.38]	-48.26 [-5.87]	2.78 [0.23]
umd	2.27 [0.89]	0.45 [0.18]	2.81 [0.97]	-3.71 [-1.43]	0.32 [0.13]	0.95 [0.36]	1.40 [0.47]
# months	684	684	504	680	684	684	500
$\bar{R}^2(\%)$	43	43	41	43	41	38	46

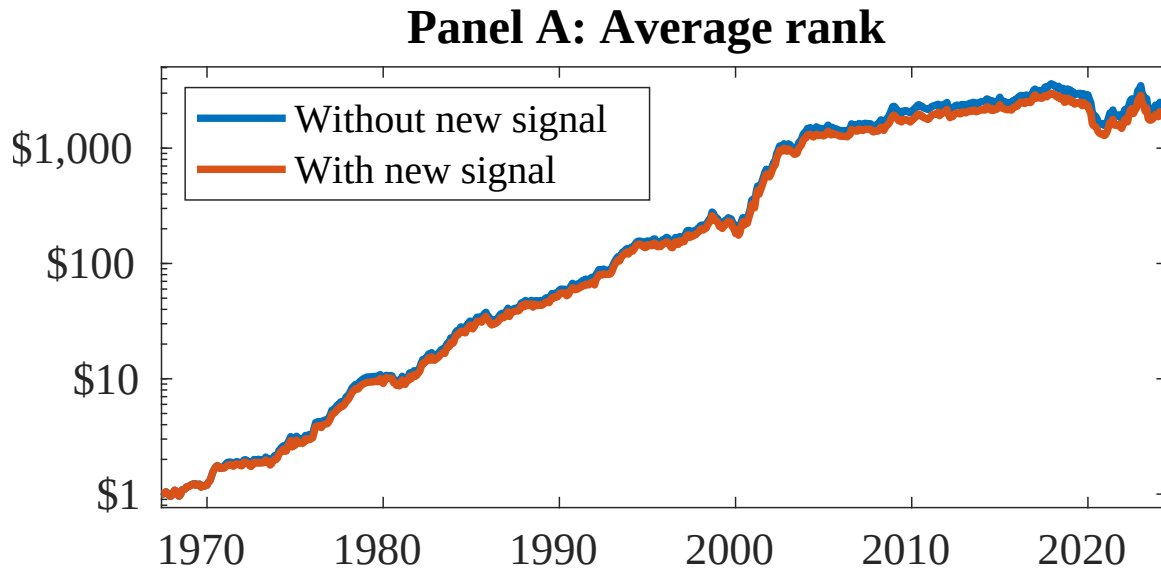


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as TPB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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